Hauke Sandhaus

HCI RESEARCHER
& UX TECHNOLOGIST

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Designing for user and worker autonomy, imagining AI systems that not just respect but support human autonomy. Dissertation work develops theory that distinguishes legitimate machine influence from manipulation.

CURRENTLY

Cornell Tech / Ph.D. Candidate in Information Science
STARTED AUGUST 2021 · EXPECTED GRADUATION JULY 2027 · NYC, NY, USA
Advised by Helen Nissenbaum (Chair) and Wendy Ju (Co-Chair). Committee: Daniel Susser (Cornell University, Ithaca) and Colin M. Gray (external, Indiana University Bloomington).

PROFESSIONAL EXPERIENCE

Cornell Tech / Graduate Research Assistant — Obfuscation Repository
SEPTEMBER 2026 – PRESENT
With Helen Nissenbaum. Building obfuscation.tech.cornell.edu, an open repository of obfuscation strategies that teaches lay users to protect their data under pervasive surveillance and AI scraping, starting from the premise that the communication channel cannot be trusted.

Center for Responsible AI at NYU / Research Assistant
MARCH 2023 – DECEMBER 2023
Worked on the 'University of Notre Dame - IBM' Technology Ethics Lab Grant Auditing AI Initiative on the project 'Expanding AI Audits to Include Instruments: Accountability, Measurements, and Data in Motion Capture Technology.'
- Collaborated with researchers from University of Michigan, Intel Labs, and New York University on an award-winning CHI publication
- Investigated bias in measurement standards for AI hardware
- Contributed to frameworks for auditing AI technologies

Cornell Tech / Graduate Research Assistant — Designer Decision-Making
JANUARY 2023 – MAY 2023
With Qian Yang. One-semester grant on external pressures shaping designer decision-making, with a mental-health-technology focus.

Cornell Tech / Graduate Research Assistant — Human–AV Interaction
JANUARY 2022 – AUGUST 2023
With Qian Yang. On-road studies of human–autonomous vehicle interaction and analysis of safety-critical data-sharing practices among AV companies. Fed into CSCW '25 and AutoUI/IEEE ITSC '24 publications.

Volkswagen Group Future Center / UX Technologist
OCTOBER 2018 – AUGUST 2021
Led the design of inclusive user experiences for fully autonomous vehicles and developed next-generation driver interfaces. Involved in the creation and transformation of CARIAD.
- Improved User Experience Questionnaire (UEQ) benchmarks by 30% over state-of-the-art through inclusive design approaches
- Developed and implemented a next-generation digital cockpit that improved accessibility of steering wheel functions without affecting driving performance
- Created high-fidelity AI prototypes using cutting-edge research papers and custom computer vision pipelines that were adopted for production development
- Conducted on-road studies for ambient in-cabin lighting systems that identified critical safety, comfort, and motion sickness factors for AV passengers
- Designed and prototyped LLM-based digital assistant interfaces for in-vehicle use, now in final stages before market release
- Established GDPR-compliant research practices that were adopted across the VW Group
- Successfully transferred multiple vision projects to strategy teams, with implementations adopted by Porsche, Skoda, and Audi

Robert Bosch / Intern & External Master Theses
AUGUST 2017 – JANUARY 2018
Designed V2X interfaces and researched interfaces for autonomous vehicles tailored for older adults and disabled users.
- Conducted research on autonomous vehicle interfaces for older adults and people with disabilities
- Designed Vehicle-to-Everything (V2X) interfaces for future mobility solutions

ELIXIER / Student Researcher
AUGUST 2016 – APRIL 2017
Conducted usability testing of multimodal and augmented workbenches for educational purposes and developed an AR manual for chemistry experiments.

100%FAT / External Bachelor Theses
JANUARY 2015 – MAY 2015

Designed, developed, and evaluated a prototype for a touchless interactive showcase in the DesignLab Twente. The prototype was permanently installed in the Design Hall.

Twente Academy / Student Teacher
MARCH 2014 – SEPTEMBER 2014
Taught upper secondary school pupils programming using the Arduino microprocessor.

EDUCATION

Cornell University, Ann S. Bowers College of Computing and Information Science / M.Sc.
2021 – 2024
Information Science
Passed Qualification Exam on October 2nd, 2024
- A Exam Title: 'Navigating Ethical Design Under External Pressures'
- Dissertation Topic: 'Empowering technology designers to create ethical technologies and resist external pressures.'

Bauhaus Universität Weimar / M.Sc.
2015 – 2018
Human-Computer Interaction
Graduated with honors (GPA 3.9)
- Master Thesis: 'An Ambient Display for Fully Autonomous Vehicles'
- Best Thesis of the Year
- Publication: 'A WOZ Study of Feedforward Information on an Ambient Display in Autonomous Cars' in The 31st Annual ACM Symposium on User Interface Software and Technology

Universiteit Twente / B.Sc.
2011 – 2015
Creative Technology
Cum Laude (Thesis GPA 4.0)
- Bachelor Thesis: 'Development of a 3D Navigable Interface for a Touchless Showcase'
- Represented graduation year and gave graduation speech
- Prototype permanently installed in the Design Hall

Czech Technical University / Semester Abroad
2013 – 2014
Electrical Engineering and Telecommunications

Gymnasium im Schloss / Abitur
2003 – 2011
Science Branch (STEM)
- Graduated with majors in Art & Religion
- Participated in and won science competitions

SERVICE & TEACHING

Course Development, Deep Learning (CS 5787) — Cornell Tech
AUGUST 2026 · NEW YORK CITY
Prepared class material with Hadar Averbuch-Elor: revised assignments and homework for student use of generative AI tools, and built a pipeline that generates final-exam questions resistant to AI-assisted answers.

Hybrid Chair — CHIWORK 2026
FALL 2025 – SUMMER 2026 · LINZ, AUSTRIA
Chairing hybrid participation and infrastructure for the 5th Annual Symposium on Human-Computer Interaction for Work.

Instructor, Ethical Vibe Coding — Cornell Tech
SUMMER 2026 · NEW YORK CITY
Three-week course on ethical AI-assisted coding for high school students at Cornell Tech's Summer Innovation Intensives. With Jonathan Segal and Wendy Ju.

Co-Instructor, Conscientious Tech Design Workshop — Cornell Tech
SUMMER 2025, SUMMER 2026 · NEW YORK CITY
Co-designed and co-facilitated an 8-week ethics workshop for Siegel PiTech PhD Impact Fellows using the Values at Play framework; operationalized the framework into collaborative whiteboard activities and assignments (pitechethics.github.io). With Helen Nissenbaum.

Teaching Assistant in UX & User Research — Cornell Tech
SPRING 2026 · NEW YORK CITY
Assisted in teaching a master's level course focused on the UX lifecycle and user research methods. With Shiri Azenkot.

Co-host NYC Privacy Day — Cornell Tech
FALL 2025 · NEW YORK CITY
Managed the website, sign-ups, and the poster and demo track for NYC Privacy Day, an initiative between privacy research institutes in the NYC area.

Teaching Assistant in Interactive Device Design — Cornell Tech
FALL 2023, FALL 2025 · NEW YORK CITY
Head TA in master level prototyping class. Maintained project code repository, labs, and oversaw grading. Updated the class for AI-codesign of technology; in Fall 2025 built WendyTA, an LLM-based teaching assistant for student projects. With Wendy Ju.

Teaching Assistant in Privacy in the Digital Age — Cornell Tech
FALL 2024 · NEW YORK CITY
Head TA in master level privacy class. Led assignments, in-class privacy policy exercise and a lecture on usable privacy. With Helen Nissenbaum and Vitaly Shmatikov.

ISGSA Liaison — Cornell University
FALL 2022 – FALL 2023 · NEW YORK CITY
Information Science Graduate Student Association Liaison for NYC to Ithaca campus collaboration.

Teaching Assistant in HCI Design — Cornell University
FALL 2021 · ITHACA
Assisted in teaching undergraduate and master level course in human computer interaction. Oversaw grading and individual sections. With Gilly Leshed.

Member — We Drive Proud
2019 – 2021 · POTSDAM, GERMANY
Member of the VW network to advocate being out at work.

Reviewer — Peer Review
2018 – ONGOING
Peer reviewer at FAccT, CHI, DIS, AutoUI, CSCW, MuC, and CHIwork.

FELLOWSHIPS & AWARDS

Public Interest Responsible Technology Fund
JUNE 2026 · CORNELL TECH
Teaching grant to instruct a PhD-fellow cohort on value-centered design methods.

Backslash Microgrant
MAY 2025 · BACKSLASH AT CORNELL TECH
Art and technology microgrant for the interactive installation 'App Design as a Commodity', a receipt printer that sells AI-built apps for Venmo change.

DLI Doctoral Fellow
2023-2024 · CORNELL TECH
The Digital Life Initiative supports examining the complexities and tensions present in digital technologies involving ethics, policy, politics, and quality of life.

Siegel Public Interest Technology Impact Fellowship
MARCH 2021 · CORNELL TECH
Funded to work for not-for-profit organization YAI. YAI is one of the largest organizations in the U.S. providing children and adults with intellectual and developmental disabilities (I/DD) a comprehensive range of services.

Thesis of the Year
JULY 2015 · UNIVERSITEIT TWENTE
Selected by jury to speak at graduation ceremony. Bachelor thesis presented on behalf of the creative technology program.

PUBLICATIONS

UNDER REVIEW · 4

- **Hauke Sandhaus**, Doris Maria Rhomberg, Helen Nissenbaum (September 2026). Persuading Metrics: How User Evaluation Data Shapes Professional Designers' Willingness to Reject Dark Patterns. *Under review, CHI '27*.
- **Hauke Sandhaus**, Jen Semler, Helen Nissenbaum, Qian Yang (September 2026). Autonomy by Design: Co-Designing a Framework to Evaluate Whether AI Interfaces Support User Autonomy. *Under review, CHI '27. Companion artifact: human-ai-autonomy-rubric.surge.sh*.
- **Hauke Sandhaus**, Ruixiang Han, Maria Teresa Parreira, Wendy Ju (September 2026). I Think We Did This: Erosion and Reassertion of Students' Design Intention in AI-Saturated Work. *Under review, CHI '27*.
- **Hauke Sandhaus**, Jae June Lee, Andrea Wang, Erica Chiang, Jennah Gosciak, Kiara Liu, Laura Greenstreet, Olzhas Yessenbayev, Helen Nissenbaum (2026). Values in Public Interest Tech: Practicing Conscientious Design. *Revise and resubmit*.

JOURNAL ARTICLES · 2 · peer-reviewed

- **Hauke Sandhaus**, Doris Maria Rhomberg (2026). Deception Is Ugly: Linking Aesthetic Judgment to Perceived Manipulation in Dark Patterns. *Zeitschrift für Psychologie*. · *Accepted, forthcoming*
- **Hauke Sandhaus**, Angel Hsing-Chi Hwang, Wendy Ju, Qian Yang (October 2025). My Precious Crash Data: Barriers and Opportunities in Encouraging Autonomous Driving Companies to Share Safety-Critical Data. *Proceedings of the ACM on Human-Computer Interaction*.

CONFERENCE PAPERS · 8 · full papers, peer-reviewed

- Emma Harvey, Emanuel Moss, **Hauke Sandhaus**, Abigail Z Jacobs, Mona Sloane (October 2026). Context and Symmetry in Auditing: A Case Study of Skeleton Inference in Motion Capture. *Proceedings of the Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES '26)*.

- **Hauke Sandhaus**, Qiuquan Gu, Maria Teresa Parreira, Wendy Ju (July 2025). Co-Designing with Transformers: Unpacking the Complex Role of GenAI in Interactive System Design Education. *Proceedings of the 2025 ACM Designing Interactive Systems Conference*.
- Matt Franchi*, **Hauke Sandhaus***, Madiha Zahrah Choksi, Severin Engelmann, Wendy Ju, Helen Nissenbaum (June 2025). Privacy of Groups in Dense Street Imagery. *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*. · * = equal contribution
- Rachel Di Pirro, **Hauke Sandhaus**, David Goedicke, Dan Calderone, Meeko Oishi, Wendy Ju (September 2024). Characterizing Cultural Differences in Naturalistic Driving Interactions. *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*.
- Navit Klein, **Hauke Sandhaus**, David Goedicke, Wendy Ju, Avi Parush (September 2024). Modeling Social Situation Awareness in Driving Interactions. *Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.
- Patrick Ebel, Pavlo Bazilinskyy, Mark Colley, Courtney Michael Goodridge, Philipp Hock, Christian P. Janssen, **Hauke Sandhaus**, Aravinda Ramakrishnan Srinivasan, Philipp Wintersberger (September 2024). Changing Lanes Toward Open Science: Openness and Transparency in Automotive User Research. *Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.
- Emma Harvey, **Hauke Sandhaus**, Abigail Z Jacobs, Emanuel Moss, Mona Sloane (May 2024). The Cadaver in the Machine: The Social Practices of Measurement and Validation in Motion Capture Technology. *Proceedings of the CHI Conference on Human Factors in Computing Systems*. · **Best Paper Honorable Mention at CHI '24**
- Aline Martinez, Michaela Honauer, **Hauke Sandhaus**, Eva Hornecker (2018). Smart Textiles in the Performing Arts. *Textiles, Identity and Innovation: Design the Future*.

SHORT PAPERS, POSTERS & LATE-BREAKING WORK · 5 · peer-reviewed

- **Hauke Sandhaus**, Doris Maria Rhomberg, Helen Nissenbaum (June 2026). Making Indecent Persuasion Visible: How Evaluation Metrics Shape UX Designers' Ethical Reasoning. *CHIWORK '26: Proceedings of the 5th Annual Symposium on Human-Computer Interaction for Work (Late-Breaking Work)*. · **Best Late-Breaking Work Award at CHIWORK '26 (1 of 25)**
- **Hauke Sandhaus**, Madiha Zahrah Choksi, Wendy Ju (September 2024). Regaining Trust: Impact of Transparent User Interface Design on Acceptance of Camera-Based In-Car Health Monitoring Systems. *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.
- Jieun Kim*, **Hauke Sandhaus***, Susan R Fussell (October 2023). VR Job Interview Using a Gender-Swapped Avatar. *Computer Supported Cooperative Work and Social Computing*. · * = equal contribution
- Tamara von Sawitzky, Ina Marie Koniakowsky, Qidan Yan, **Hauke Sandhaus**, Ali Askari (September 2023). Exploring Interactions for Collaborative Driving in Fully Automated Vehicles. *Adjunct Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.
- **Hauke Sandhaus**, Eva Hornecker (October 2018). A WOZ Study of Feedforward Information on an Ambient Display in Autonomous Cars. *The 31st Annual ACM Symposium on User Interface Software and Technology Adjunct Proceedings*.

WORKSHOP PAPERS · 5

- Maria Teresa Parreira, Pranav Prabhat Sinha, **Hauke Sandhaus**, Wendy Ju (June 2026). Visible Adoption, Untracked Contribution: GitHub Evidence of the Accountability Gap Across Three Cohorts of an HCI Prototyping Course. *CHIWORK '26: Adjunct Proceedings of the 5th Annual Symposium on Human-Computer Interaction for Work*. · *Workshop paper at CHIWORK '26 (workshop I co-organized)*
- **Hauke Sandhaus**, Maria Teresa Parreira, Wendy Ju (May 2024). Student Reflections on Self-Initiated GenAI Use in HCI Education. *LLMs as Research Tools Workshop at the CHI Conference on Human Factors in Computing Systems*.
- Doris Maria Rhomberg*, **Hauke Sandhaus*** (May 2024). Towards Quantifying Ethical User Experience: Evaluating User Perceptions of Dark Patterns in Social Media. *Mobilizing Research and Regulatory Action on Dark Patterns and Deceptive Design Practices Workshop at the CHI Conference on Human Factors in Computing Systems*. · * = equal contribution
- **Hauke Sandhaus** (April 2023). Promoting Bright Patterns. *CHI '23 Workshop: Designing Technology and Policy Simultaneously*. · *This paper started the brightpatterns.org website*
- **Hauke Sandhaus**, Wendy Ju, Qian Yang (April 2023). Towards Prototyping Driverless Vehicle Behaviors, City Design, and Policies Simultaneously. *CHI '23 Workshop: Designing Technology and Policy Simultaneously*.

WORKSHOPS ORGANIZED · 5 · as co-organizer

- Kashif Imteyaz, Mohammad Rashidujjaman Rifat, Divya Ramesh, Steven R Rick, Simo Hosio, **Hauke Sandhaus**, Advait Sarkar, Christoph Riedl, Saiph Savage (November 2026). "Nobody Did This": Contribution, Originality, and Accountability in Agent-Mediated Collaboration. *Proceedings of the 2026 Companion of the Computer-Supported Cooperative Work and Social Computing*. · *Workshop co-organizer*
- **Hauke Sandhaus**, Kashif Imteyaz, Mohammed Almutairi, Pooja Prajod, Divya Ramesh, Saiph Savage, Qian Yang, Michael Muller (June 2026). Interrogating GenAI Augmentation for CHIworkers: Strategies for Professional Autonomy and Accountability. *CHIWORK '26: Proceedings of the 5th Annual Symposium on Human-Computer Interaction for Work (Workshop)*. · *Workshop co-organizer*
- Thomas Eßmeyer, Johanna Gunawan, Colin M Gray, Lorena Sánchez Chamorro, Alberto Monge Roffarello, Katie Seaborn, **Hauke Sandhaus**, Kai Lukoff, Gian-Luca Savino, Shruthi Sai Chivukula, Satoshi Nakamura (April 2026). Bridge Over Troubled Water: Aligning Commercial Incentives With Ethical Design Practice To Combat Deceptive Patterns. *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. · *Workshop co-organizer*
- Colin M Gray, Johanna T Gunawan, René Schäfer, Nataliia Bielova, Lorena Sánchez Chamorro, Katie Seaborn, Thomas Mildner, **Hauke Sandhaus** (May 2024). Mobilizing Research and Regulatory Action on Dark Patterns and Deceptive Design Practices. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. · *Workshop co-organizer; co-editor of the workshop proceedings, CEUR-WS Vol. 3720*
- Patrick Ebel, Pavlo Bazilinskyy, Angel Hsing-Chi Hwang, Wendy Ju, **Hauke Sandhaus**, Aravinda Ramakrishnan Srinivasan, Qian Yang, Philipp Wintersberger (September 2023). Breaking Barriers: Workshop on Open Data Practices in AutoUI Research. *Adjunct Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. · *Workshop co-organizer*

OTHER · 2 · thesis, non-archival

- **Hauke Sandhaus**, Helen Nissenbaum (October 2025). The Privacy Policy Analysis Canvas: An Ethics Thinking Activity for Teaching Privacy Policy Insufficiency. *Poster at NYC Privacy Day 2025, Cornell Tech, New York, NY*. · *Non-archival poster; write-up unpublished*
- **Hauke Sandhaus** (2015). Development of a 3D Navigable Interface for a Touchless Showcase. *Bachelor's thesis, University of Twente*.

SKILLS

Coding: Python, Java, JavaScript, React.js, Node.js, Android, Unity3D, C#, Arduino, Open Frameworks
AI: fastai, Pinecone, Teachable Machines, TensorFlow, ml5.js, OpenCV, Scikit-learn, DepthAI, LLM integration, Custom computer vision pipelines
Data Analytics: Python data visualization, Tableau, R, Matlab, d3.js
Design: Interface design, Lottie, AE, 3D modeling, PSD, Framer
User Research: Contextual & expert interviews, Ethnography, Surveys, Usability testing, On-road studies
UX: Personas, Wireframes, Focus groups, Human factors, Inclusive design, Accessibility
Hardware: CAD, Rapid Prototyping, 3D manufacturing
Management: Team leadership, Conflict resolution, Strategy, Scaled agile methods (certified), UX strategy

REFERENCES

Helen Nissenbaum
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Tisch Professor of Information Science · Cornell Tech
Ph.D. advisor (Chair); Director, Digital Life Initiative

Wendy Ju wendyju@cornell.edu
Associate Professor of Information Science · Cornell Tech
Ph.D. co-advisor (Co-Chair)

Colin M. Gray comgray@iu.edu
Professor of Informatics · Indiana University Bloomington
External committee member

Daniel Susser
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Associate Professor of Information Science · Cornell University, Ithaca
Committee member

Mona Sloane
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Assistant Professor of Data Science and Media Studies · University of Virginia
Supervisor at the NYU Center for Responsible AI, 2023; co-author

Eva Hornecker ehornecker.de
Professor of Human-Computer Interaction · Bauhaus-Universität Weimar
M.Sc. thesis advisor

Edwin Dertien edwindertien.nl
Assistant Professor, Robotics and Mechatronics; Creative Technology · University of Twente
B.Sc. thesis supervisor

Klaus Zyciora Wikipedia
President, Changan Automobile (global design) · formerly Head of Volkswagen Group Design
Worked together at the Future Center Europe in 2019

Ulrike Müller autocontact.at
Former Managing Director, VW Group Future Center Europe
Worked together during my UX Technologist years, 2018–2021

Contact details for the remaining references available on request.